

An Oscillatory Shear Excludes the Lipschitz Approximation Endpoint

fa_banach_001 / agent_lane_14

13 August 2026

Status: candidate full counterexample, likely valid; human review needed. The endpoint 1 does not belong to the approximation set $B(1, 1)$ of Bellido–Mora-Corral. In fact, one explicit bi-Lipschitz shear stays Lipschitz distance at least one from every map affine on a finite triangulation. Together with the source theorem, this gives $B(1, 1) = (0, 1)$.

1 Source question

Bellido and Mora-Corral study approximation of planar Hölder homeomorphisms by piecewise affine homeomorphisms over Euclidean finite complexes [1]. For $0 < \alpha, \tilde{\alpha} \leq 1$, they let $B(\alpha, \tilde{\alpha})$ be the set of exponents $0 < \beta \leq 1$ for which, on every closed polygon Ω , piecewise affine homeomorphisms are dense in

$$\{h \in C^\alpha(\Omega, \mathbb{R}^2) : h \text{ is a homeomorphism and } h^{-1} \in C^{\tilde{\alpha}}(h(\Omega), \mathbb{R}^2)\}$$

in the C^β norm. Their Theorem 7.6 implies $\sup B(1, 1) = 1$. On PDF page 45 they ask whether the endpoint belongs:

$$1 \stackrel{?}{\in} B(1, 1).$$

Approximation of Hölder continuous homeomorphisms

45

(the latter equality comes from (97), (73), (58) and (2)). An optimal version of the result of Theorem 7.6 would be to calculate $\sup B(\alpha, \tilde{\alpha})$ and to ascertain whether $\sup B(\alpha, \tilde{\alpha})$ belongs to $B(\alpha, \tilde{\alpha})$. For example, it is easy to show that

$$\alpha \notin B(\alpha, \tilde{\alpha}), \quad \alpha < 1, \quad \tilde{\alpha} \leq 1. \quad (111)$$

Indeed, take $0 < \alpha < 1$, and let Ω be a closed polygon whose interior contains 0. Define $h : \Omega \rightarrow \mathbb{R}^2$ by

$$h(x, y) = (|x|^\alpha \operatorname{sgn} x, y), \quad (x, y) \in \Omega,$$

where $\operatorname{sgn} x$ stands for the sign of $x \in \mathbb{R}$. Then $h \in C^\alpha(\Omega, \mathbb{R}^2)$ is a homeomorphism, and h^{-1} is Lipschitz continuous. However, h cannot be approximated by piecewise affine homeomorphisms in the $\|\cdot\|_\alpha$ norm. In fact, it is easy to see that h cannot be approximated by Lipschitz continuous functions in the $\|\cdot\|_\alpha$ norm (see, e.g., Kichenassamy [15], if necessary). Therefore, $\alpha \notin B(\alpha, 1)$, and this implies (111).

Note that Theorem 7.6 shows in particular that $\sup B(1, 1) = 1$, but we do not know whether 1 belongs to $B(1, 1)$. In any case, we believe that Theorem 7.6 is not optimal.

Figure 1: The endpoint question on PDF page 45 of arXiv:0806.3366.

2 Definitions

For a map F on the unit square $Q = [0, 1]^2$, write

$$|F|_1 = \sup_{p \neq q \in Q} \frac{|F(p) - F(q)|}{|p - q|}, \quad \|F\|_1 = \|F\|_\infty + |F|_1.$$

This is the source's exponent-one Hölder norm. Let $\text{PA}_{\text{fin}}(Q, \mathbb{R}^2)$ denote all maps affine on the triangles of some finite triangulation of Q ; injectivity is not imposed in this larger class.

3 Proof intuition

The endpoint norm detects slopes rather than only function values. A finite piecewise affine map has only finitely many slopes along each boundary edge, so near a corner its boundary slope is constant. We build a horizontal shear whose boundary slope continues to alternate between -1 and $+1$ at arbitrarily small scales. The shear and its inverse remain Lipschitz because the slope is bounded. No constant can be within less than one of both -1 and $+1$, which yields a uniform gap from every finite piecewise affine map.

4 Counterexample and exact endpoint

Define $g : [0, 1] \rightarrow \mathbb{R}$ by

$$g(0) = 0, \quad g(t) = t^2 \sin(1/t) \quad (t > 0),$$

and define

$$h : Q \longrightarrow \mathbb{R}^2, \quad h(x, y) = (x + g(y), y). \quad (1)$$

Theorem 1. *The map h in (1) is a bi-Lipschitz homeomorphism from Q onto $h(Q)$. Moreover, for every $f \in \text{PA}_{\text{fin}}(Q, \mathbb{R}^2)$,*

$$\|h - f\|_1 \geq |h - f|_1 \geq 1. \quad (2)$$

Consequently, $1 \notin B(1, 1)$.

Proof. For $t > 0$,

$$g'(t) = 2t \sin(1/t) - \cos(1/t), \quad (3)$$

so $|g'(t)| \leq 3$ on $(0, 1]$. Also $g'(0) = 0$ directly from the difference quotient. Thus g is Lipschitz. The shear is injective because it preserves the second coordinate, and its inverse on $h(Q)$ is

$$h^{-1}(u, v) = (u - g(v), v).$$

The triangle inequality and the Lipschitz bound for g show that both maps are Lipschitz. Hence h satisfies the defining regularity hypotheses for $B(1, 1)$.

Fix $f \in \text{PA}_{\text{fin}}(Q, \mathbb{R}^2)$ and restrict it to the left edge $E = \{(0, t) : 0 \leq t \leq 1\}$. Intersecting a finite triangulation with E gives a finite subdivision of E , so there are $\delta > 0$ and $a, c \in \mathbb{R}$ such that

$$\pi_1 f(0, t) = a + ct \quad (0 \leq t \leq \delta).$$

The first coordinate of $(h - f)(0, t)$ on this interval is therefore $d(t) = g(t) - a - ct$. For all sufficiently large n , both

$$t_n = \frac{1}{2\pi n}, \quad s_n = \frac{1}{(2n+1)\pi}$$

belong to $(0, \delta)$. Equation (3) gives exactly

$$g'(t_n) = -1, \quad g'(s_n) = 1.$$

The absolute value of a derivative at any differentiability point is bounded by the Lipschitz seminorm. Consequently,

$$\begin{aligned} |h - f|_1 &\geq \text{Lip}(d; [0, \delta]) \\ &\geq \max\{|-1 - c|, |1 - c|\} = 1 + |c| \geq 1, \end{aligned}$$

which proves (2). Since this excludes approximation even by the larger class $\text{PA}_{\text{fin}}(Q, \mathbb{R}^2)$, it certainly excludes finite piecewise affine homeomorphisms. The square is a closed polygon, so the universal density property defining $1 \in B(1, 1)$ fails. $\square$

Corollary 1. *The source approximation set is exactly*

$$B(1, 1) = (0, 1).$$

Proof. For $\alpha = \tilde{\alpha} = 1$, source Theorem 7.6 gives approximation in every exponent $0 < \beta < 1$, hence $(0, 1) \subseteq B(1, 1)$. Theorem 1 excludes the only remaining endpoint. $\square$

5 Verification, scope, and novelty

The included script `code/verify_shear.py` checks 3,500 instances of the two exact slope sequences against seven affine slopes and verifies the shear/inverse identities on a 101×101 grid. This is only a numerical sanity check; the proof is the elementary calculation above. Details are in `verification.md`.

The counterexample settles precisely the source’s $B(1, 1)$ endpoint. It is stronger than needed because the lower bound holds for noninjective finite piecewise affine maps as well. It makes no assertion about approximation by countably piecewise affine maps with subdivisions accumulating at the corner, nor about stronger quantitative bi-Lipschitz approximation questions.

Cheap run indexes, the parsed local corpus, and bounded web/arXiv searches through 13 August 2026 used the exact endpoint phrase, title and authors, $B(1, 1)$, finite piecewise affine density, Lipschitz-norm approximation, and oscillatory-shear terms. They found the source and later Sobolev and bi-Lipschitz approximation literature, but no explicit answer to this exact endpoint question. See `novelty_search.md`. This is not an exhaustive priority claim.

6 Human-review recommendation

The highest-value check is the finite-complex convention: the restriction of a map affine on a finite triangulation to a polygon edge must have a first affine interval. A reviewer should also confirm that the source’s $\|\cdot\|_1$ is the ordinary Lipschitz norm. Once those definitions are matched, the explicit inverse and the two derivative sequences make the counterexample immediate.

References

- [1] J. C. Bellido and C. Mora-Corral, *Approximation of Hölder continuous homeomorphisms by piecewise affine homeomorphisms*, arXiv:0806.3366 (2008).